

◆ Slide 1: Title Slide

“Good morning everyone.

My name is Yogiraj Patidar and I am here to deliver a powepoint presentation on **Cybersecurity in 6G Networks** with my team members: Deepak, Adnan and Bhavna

◆ Slide 2: Understanding 6G Technology

“6G is the next generation of wireless communication after 5G.

It promises speeds that are almost **one hundred times faster** than 5G.

It will also offer **ultra-low latency**, which means there will be almost no delay in data transmission.

With these features, 6G will support **AI-based networks** that can make decisions, analyze data, and manage operations automatically.

(You can add: Latency refers to the time taken for data to travel from one point to another. In 6G, this time becomes extremely small, which allows instant communication.)”

Now lets discuss Applications of 6G

“The arrival of 6G will bring several revolutionary applications.

For example, **fully autonomous vehicles** will be able to communicate instantly with each other and make safer decisions on the road.

Holographic communication will allow people to interact with 3D holograms as if they are physically present.

Smart cities will use real-time intelligence to improve traffic, healthcare, and public services.

There will also be **massive IoT ecosystems**, where trillions of devices can connect and communicate seamlessly.

All of these applications require extremely fast and secure networks so 6G network is important for this aspect.

◆ Slide 4: Why Cybersecurity Matters More Than Ever in 6G

“As 6G networks become more advanced, cybersecurity becomes even more important.

This is because the number of connected devices will increase dramatically(suddenly).

More devices mean more opportunities for cybercriminals to attack.

Cybercriminals are also becoming more advanced, and they use modern tools and artificial intelligence to break into systems.

At the same time, huge amounts of personal and sensitive data will be transmitted through 6G networks.

and hence, protecting this data and preventing cyber-attacks will become a very big challenge.”

◆ Slide 5: Critical Security Challenges Ahead

“There are several major security challenges in 6G networks.

First, there is an **expanded attack surface**.

This means that since many devices are connected, hackers have more entry points to attack the network.

Second, there are **data privacy concerns**.

As data moves across many devices and platforms, protecting personal information becomes more difficult.

Third, **device authentication** becomes a challenge.

It is hard to identify whether a device is real or fake, which can allow cloned devices to enter the network.

Fourth, we face **AI-powered threats**.

Hackers are now using artificial intelligence to create smarter and more adaptive attacks that are harder to detect.

Fifth, there are **IoT vulnerabilities**.

Many IoT devices do not have strong security, and even one weak device can become a point of attack for the entire network.

◆ Slide 6: Common Cyber Threats in 6G Networks

“6G networks are exposed to several common cyber threats.

One major threat is **data theft**: data theft means attackers steal personal or organisational information.

Another threat is **identity theft**: in identity theft criminals pretend to be legitimate users to gain access to systems.

There is also **system hacking**: in system hacking attackers get into networks and take control of devices or steal data.

In addition, there is **DDoS attacks**: it overloads the network with too much traffic, causing important services to stop working.

and at last there are **malware and viruses**: these are harmful programs that damage systems or give attackers access to private information.

(All these threats become more dangerous because 6G networks are faster, larger, and more interconnected.)”

◆ Slide 7: Advanced Security Technologies for 6G

“To protect 6G networks, several advanced security technologies are being developed.

Strong encryption can be used to convert data into an unreadable form so that even if it is stolen, it cannot be understood.

Secure authentication ensures that only verified users and trusted devices can enter the

network.

Blockchain technology: it helps create tamper-proof records and builds trust without needing a central authority.

Continuous monitoring means the network is observed at all times to detect unusual behaviour.

AI-based threat detection uses artificial intelligence to identify patterns, predict attacks, and take action automatically.

(You can say: These technologies work together to create a stronger and smarter security system for 6G networks.)”

◆ Slide 8: The Dual Role of Artificial Intelligence

“Artificial intelligence plays a double role in 6G cybersecurity.

On one hand, **AI acts as a defender**.

It monitors the network, detects unusual activity, predicts possible attacks, and helps prevent security breaches before they happen.

On the other hand, **AI can be used as a weapon** by cybercriminals.

Hackers use AI to create advanced malware that can learn, adapt, and bypass security systems by its own.

This creates an ongoing battle between AI used for security and AI used for attacks.

(This means that the future of cybersecurity will depend on who uses AI more effectively.)”

◆ Slide 9: How Users Can Protect Themselves

“Even though 6G networks will be advanced, users must also follow safety practices.

They should use **strong and unique passwords** to secure their accounts.

They must **regularly update their devices**, because updates fix security issues.

Users must only install **trusted applications** and avoid unknown or suspicious apps.

They should also **review their privacy settings** to control what data is being shared.

Finally, users must **stay alert online** and avoid clicking on suspicious links or downloading unknown files.

◆ Slide 10: Conclusion

“To conclude my presentation, I would like to say that 6G will bring revolutionary technology and completely transform the digital world.

However, without strong cybersecurity, 6G networks will be vulnerable to many risks.

We must focus on securing our data, devices, and networks to enjoy the full benefits of 6G safely.

Preparing for future cyber threats today will help us build a secure and trustworthy digital future.

Thank you everyone for your attention.”